EE121 Discussion 2

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Agenda

- Complex Baseband
  - Finish Problem 2.15
- Channel Delay Spread and Coherence Bandwidth
  - Example Problem
- Amplitude Modulation (AM)
  - Example Problem

Problem 2.15 Consider the signal $s(t) = I_{[-1,1]}(t)\cos(400\pi t)$.

(a) Find and sketch the baseband signal $u(t)$ that results when $s(t)$ is downconverted as shown in the upper branch of Figure 2.39.

(b) The signal $s(t)$ is passed through the bandpass filter with impulse response $h(t) = I_{[0,1]}(t)\sin(400\pi t + \pi/4)$. Find and sketch the baseband signal $v(t)$ that results when the filter output $y(t) = (s * h)(t)$ is downconverted as shown in the lower branch of Figure 2.39.

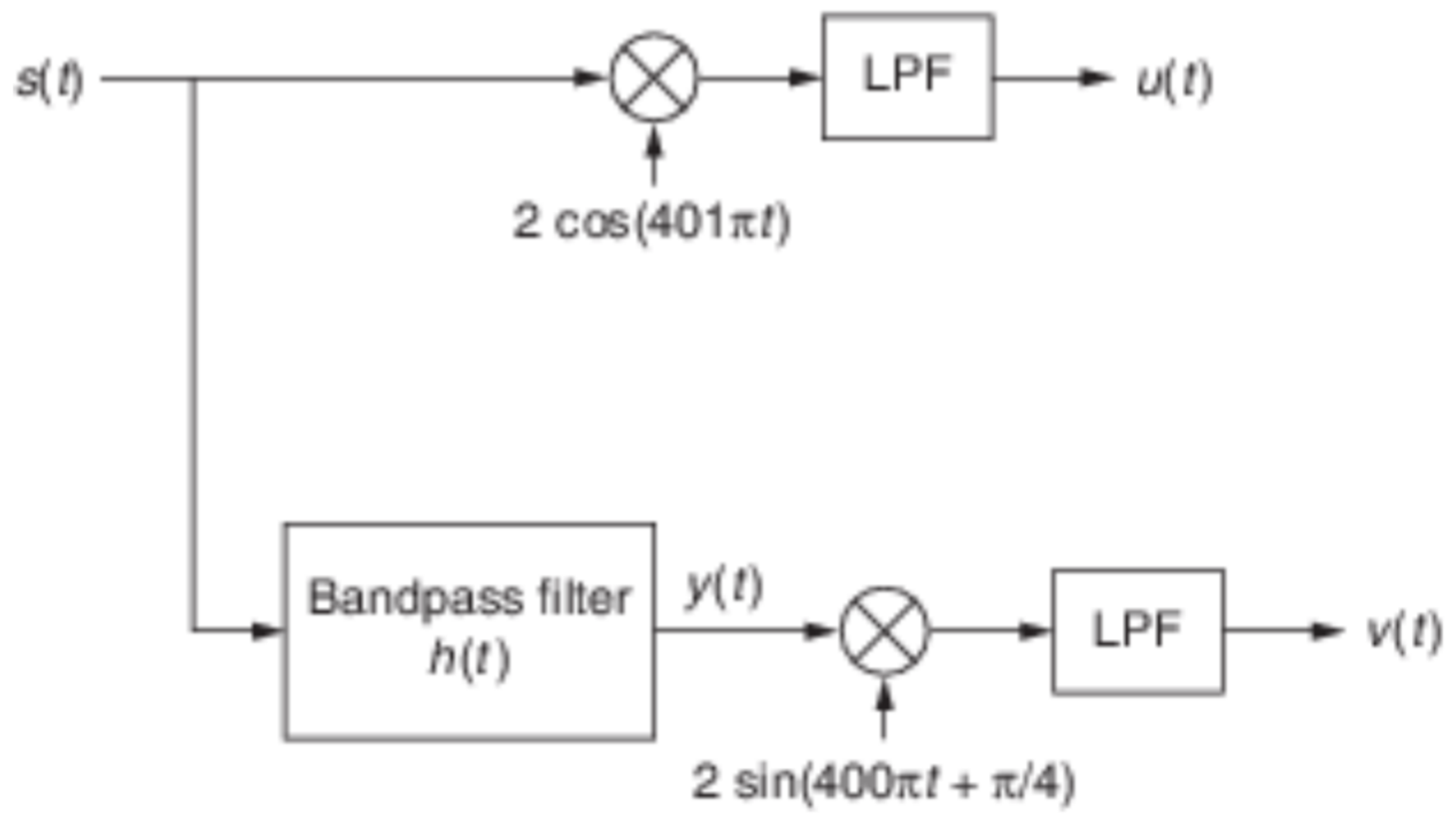
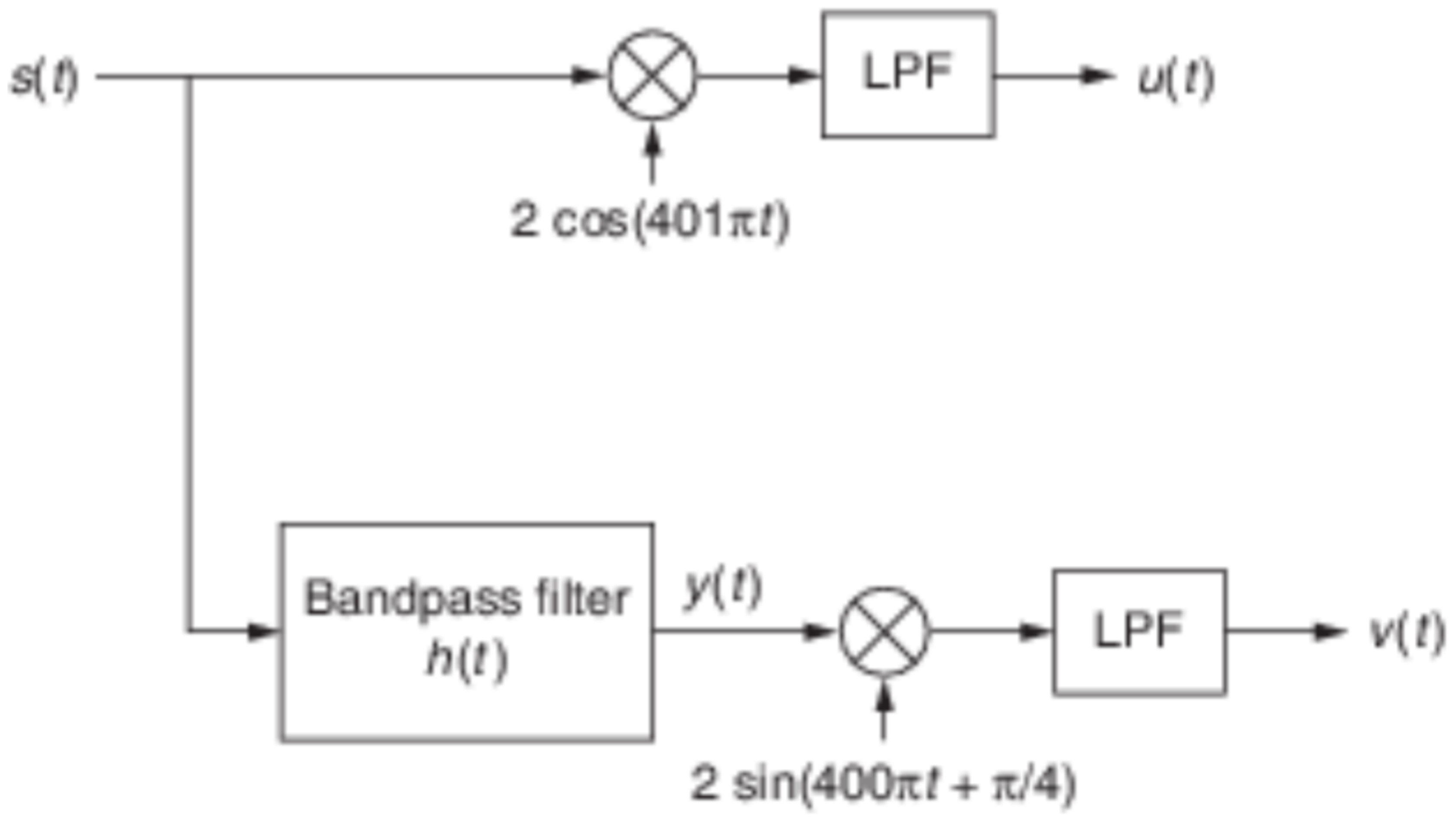
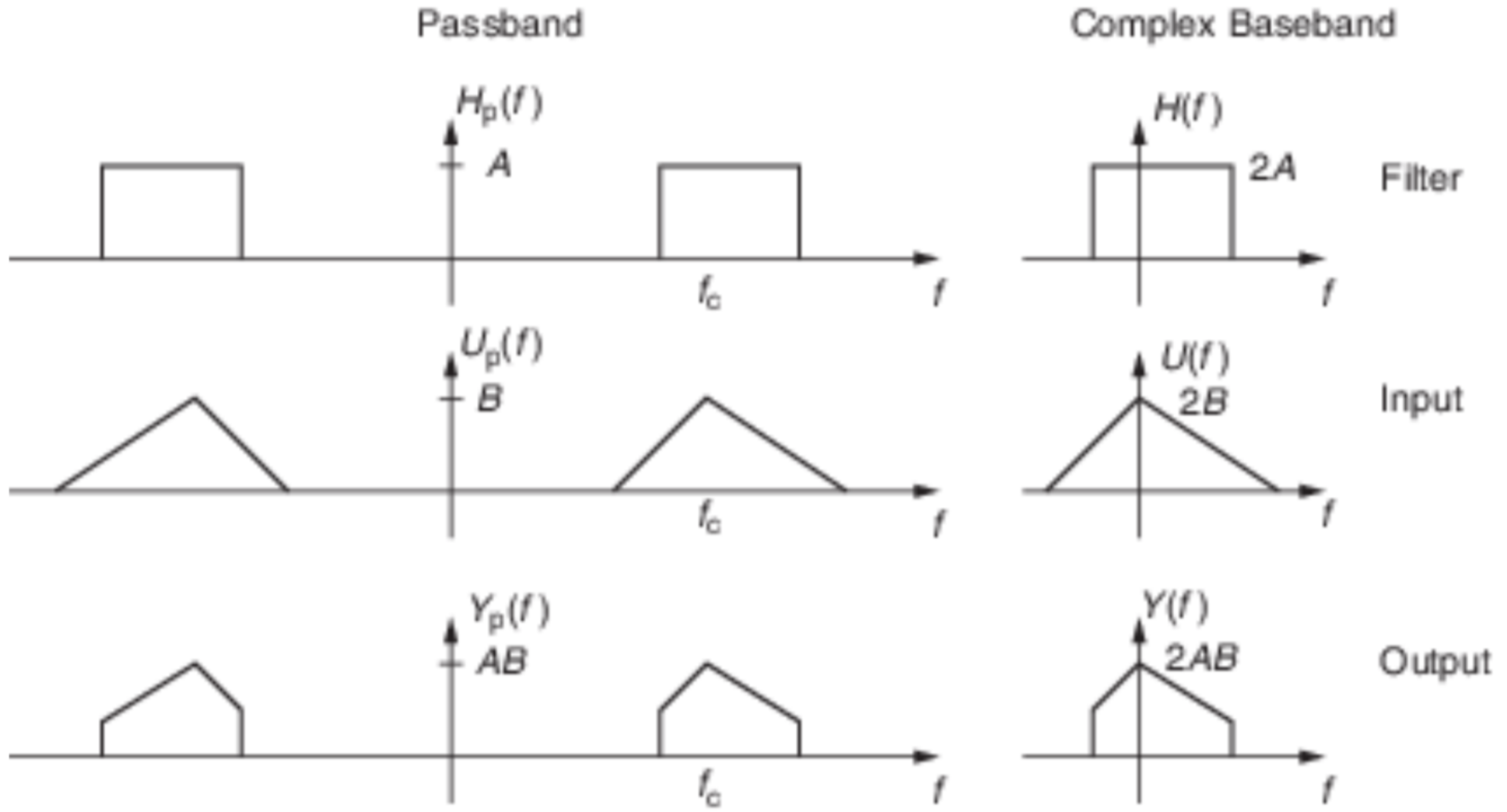


Figure 2.39 Operations involved in Problem 2.15.





$$v(t) = -y_s(t), \quad y(t) = y_c(t)\cos(400\pi t + \pi/4) - y_s(t)\sin(400\pi t + \pi/4)$$

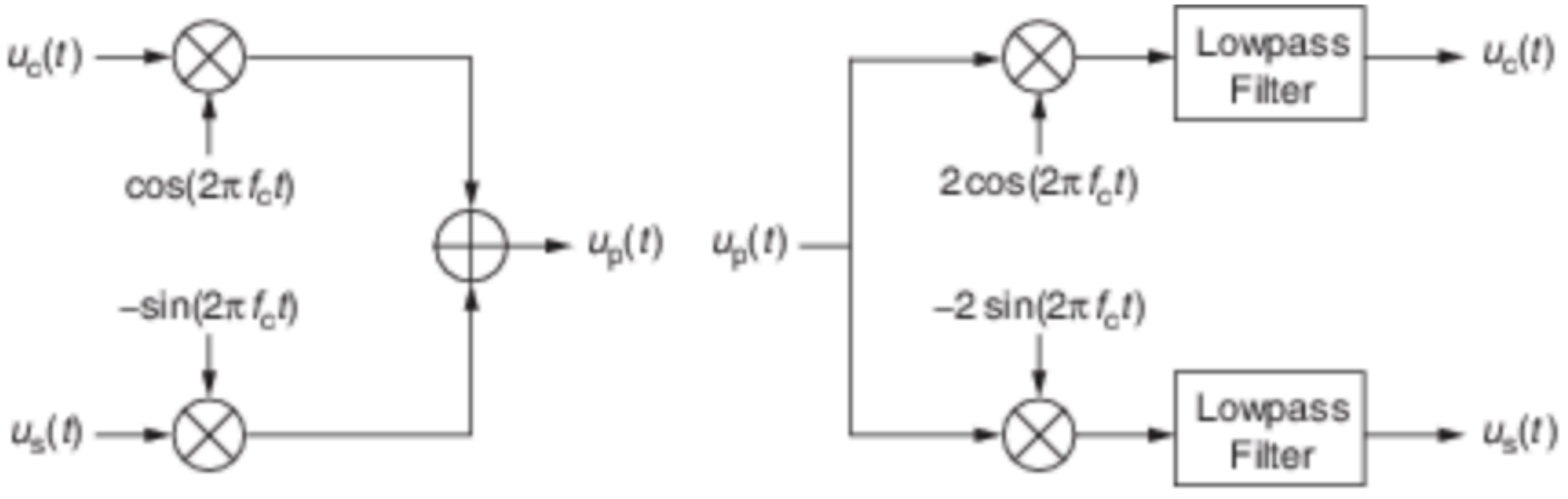
$$y(t) = \mathrm{Re}\left\{ \tilde{y}(t)\, e^{j(400\pi t + \pi/4)} \right\}$$

$$\tilde{y}(t) = \frac{1}{2}\left(\tilde{s}(t) * \tilde{h}(t)\right) = \frac{1}{2} p(t)\, e^{-j3\pi/4}$$

$$p(t) = I_{[-1,1]}(t) * I_{[0,1]}(t)$$

$$v(t) = -y_s(t) = -\mathrm{Im}\{\tilde{y}(t)\} = \frac{1}{2} p(t) \sin 3\pi/4$$

$$\boxed{v(t) = \frac{1}{2\sqrt{2}}\, p(t)}$$



Upconversion
(baseband to passband)

Downconversion
(passband to baseband)

Channel Delay Spread and Coherence Bandwidth

$$u_p(t) = u_c(t)\cos(2\pi f_c t) - u_s(t)\sin(2\pi f_c t) = e(t)\cos(2\pi f_c t + \theta(t))$$

Passband Signal

$$u(t) = u_c(t) + j u_s(t) = e(t)e^{j\theta(t)}$$

Complex Baseband (Complex Envelope)

$$v_p(t) = \frac{A}{r} e(t-\tau)\cos(2\pi f_c (t-\tau) + \theta(t-\tau) + \phi)$$

Received Signal

Find the complex envelope

$$v_p(t) = \frac{A}{r} e(t-\tau)\cos(2\pi f_c(t-\tau) + \theta(t-\tau) + \phi)$$

Received Signal

$$v(t) = \frac{A}{r} u(t-\tau) e^{-j(2\pi f_c \tau + \phi)}$$

Multiple Paths

$$v(t) = \sum_i \frac{A_i}{r_i} u(t - \tau_i) e^{-j(2\pi f_c \tau_i + \phi_i)}$$

Complex Envelope for Multiple Paths

What's the complex impulse response?

$$h(t) = \sum_i \frac{A_i}{r_i} e^{-j(2\pi f_c \tau_i + \phi_i)} \delta(t - \tau_i)$$

Channel delay spread and coherence bandwidth

Let τ_{min} and τ_{max} the minimum and maximum of the delays $\{\tau_i\}$

$\tau_d = \tau_{max} - \tau_{min}$	$B_c = 1/\tau_d$
Channel Delay Spread	Coherence Bandwidth

A baseband signal of bandwidth W is said to be narrowband if

$$W\tau_d = W/B_c < < 1$$

Look a wireless channel!

in μs

$$h(t) = 2\delta(t - 0.1) + j\delta(t - 0.64) - 0.8\delta(t - 2.2)$$

Channel Impulse Response

What is the delay spread and coherence bandwidth?

$$\tau_d = \tau_{max} - \tau_{min} = 2.1\,\mu s, \quad B_c = \frac{1}{\tau_d} = \frac{1}{2.1} = 476\,kHz$$

Look a wireless channel!

$$h(t) = 2\delta(t - 0.1) + j\delta(t - 0.64) - 0.8\delta(t - 2.2)$$

Channel Impulse Response

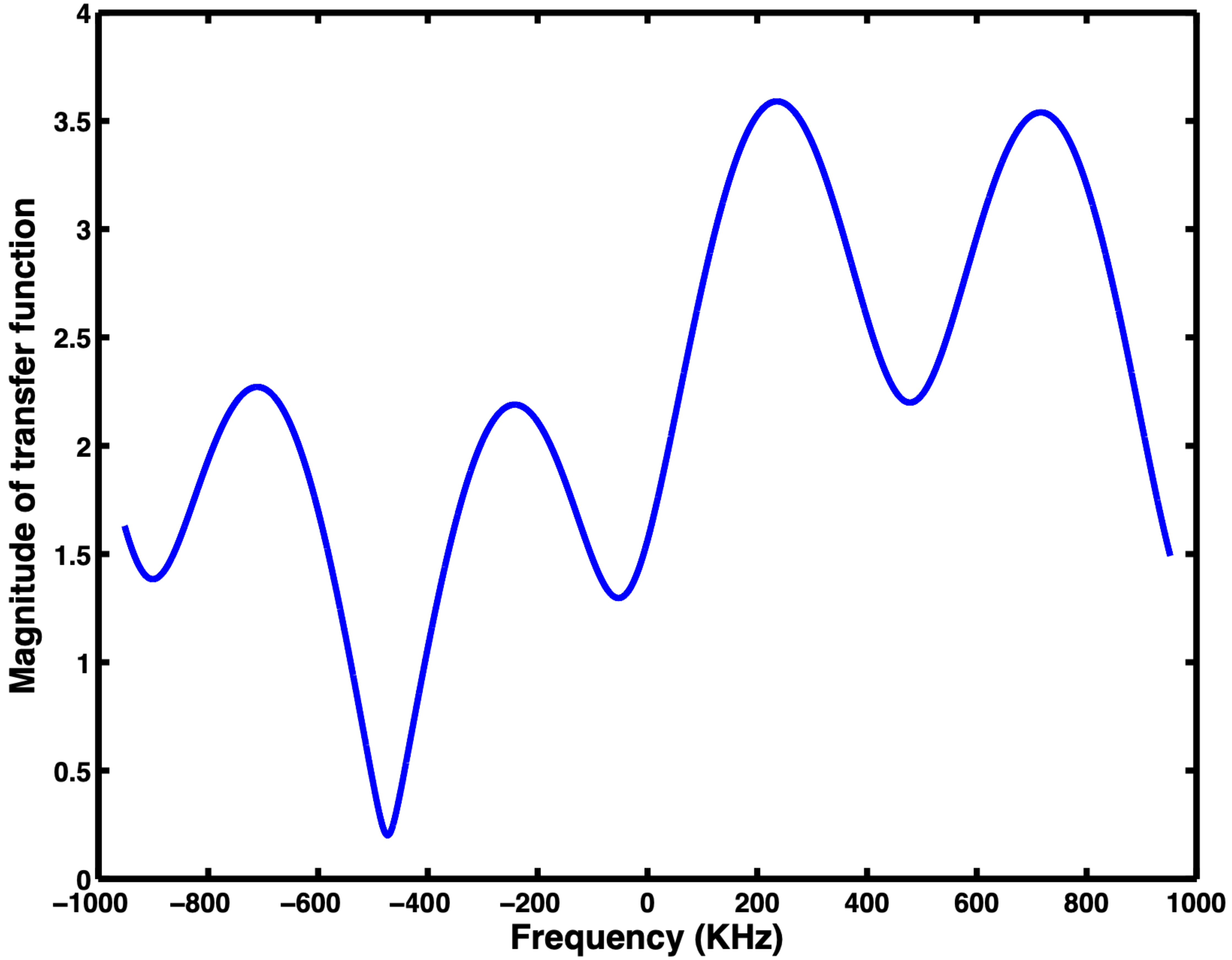
Plot the magnitude and phase of the channel transfer function $H(f)$ over the interval $[-2B_c, 2B_c]$

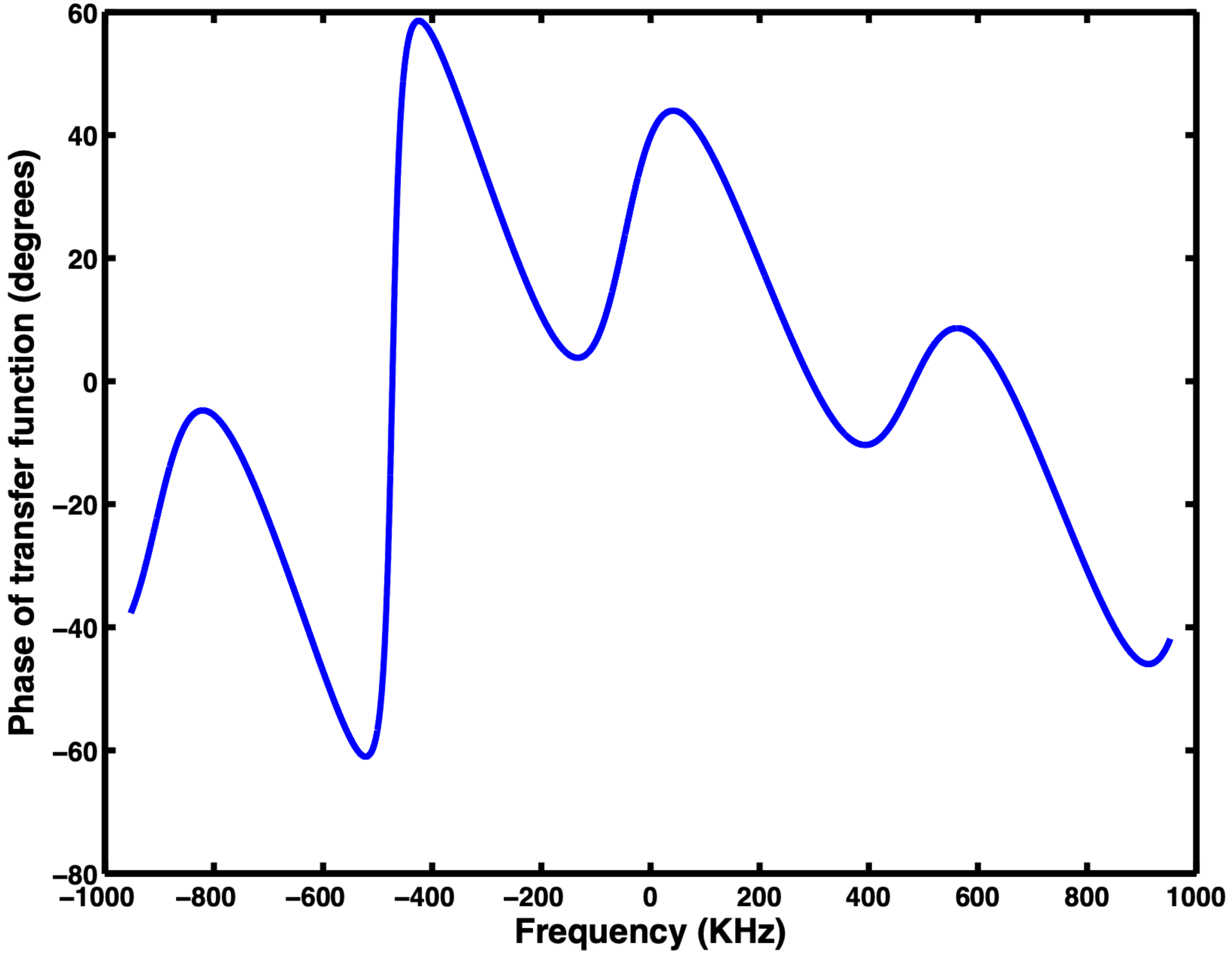
$$H(f) = 2 + je^{-j2\pi f\Delta_1} - 0.8e^{-j2\pi f\Delta_2}$$

$\Delta_1 = 0.54\mu s$, $\Delta_2 = 2.1\mu s$

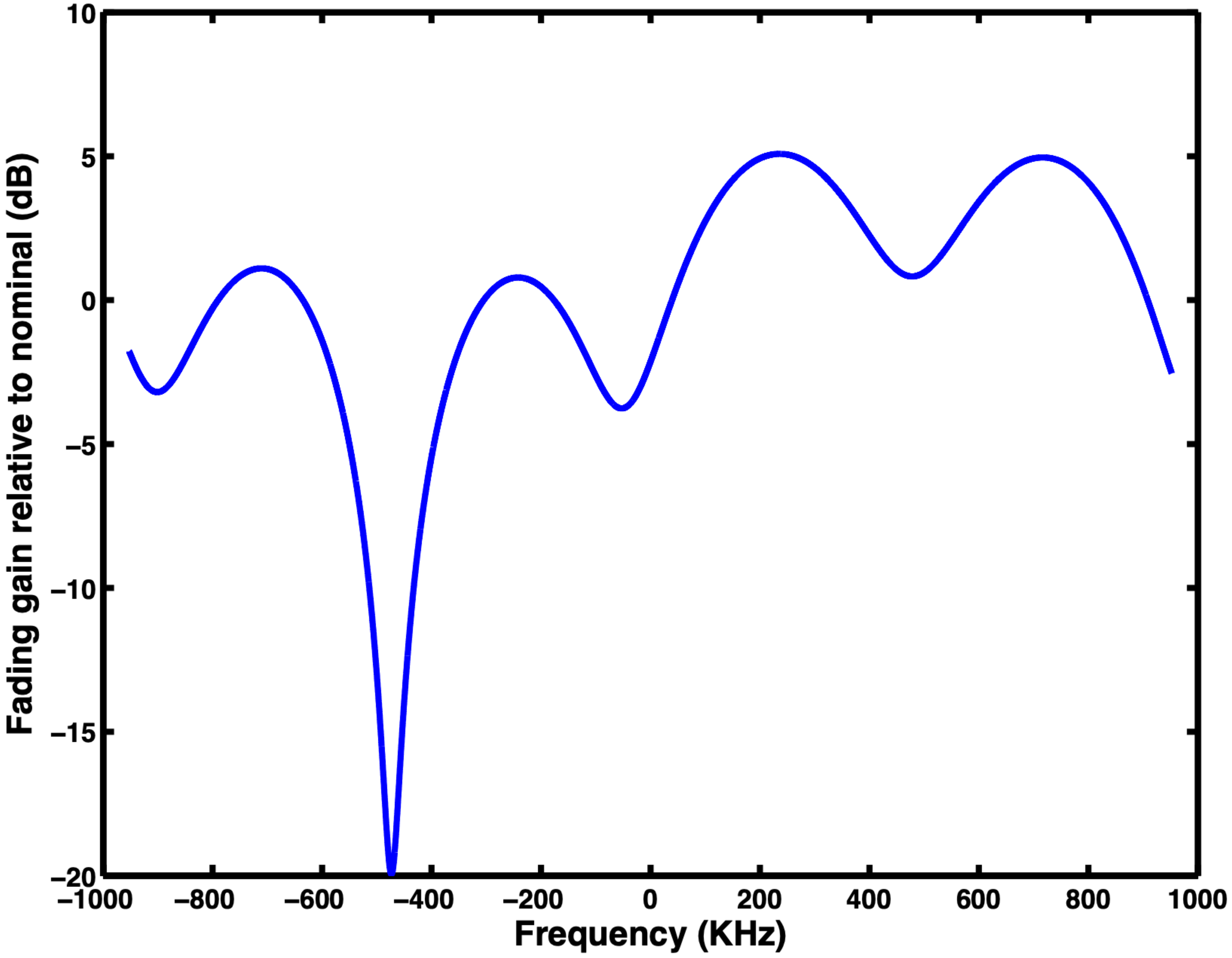
* We dropped the delay from the 1st term!

Magnitude and Phase of the Channel Transfer Function





Spectrum in dB



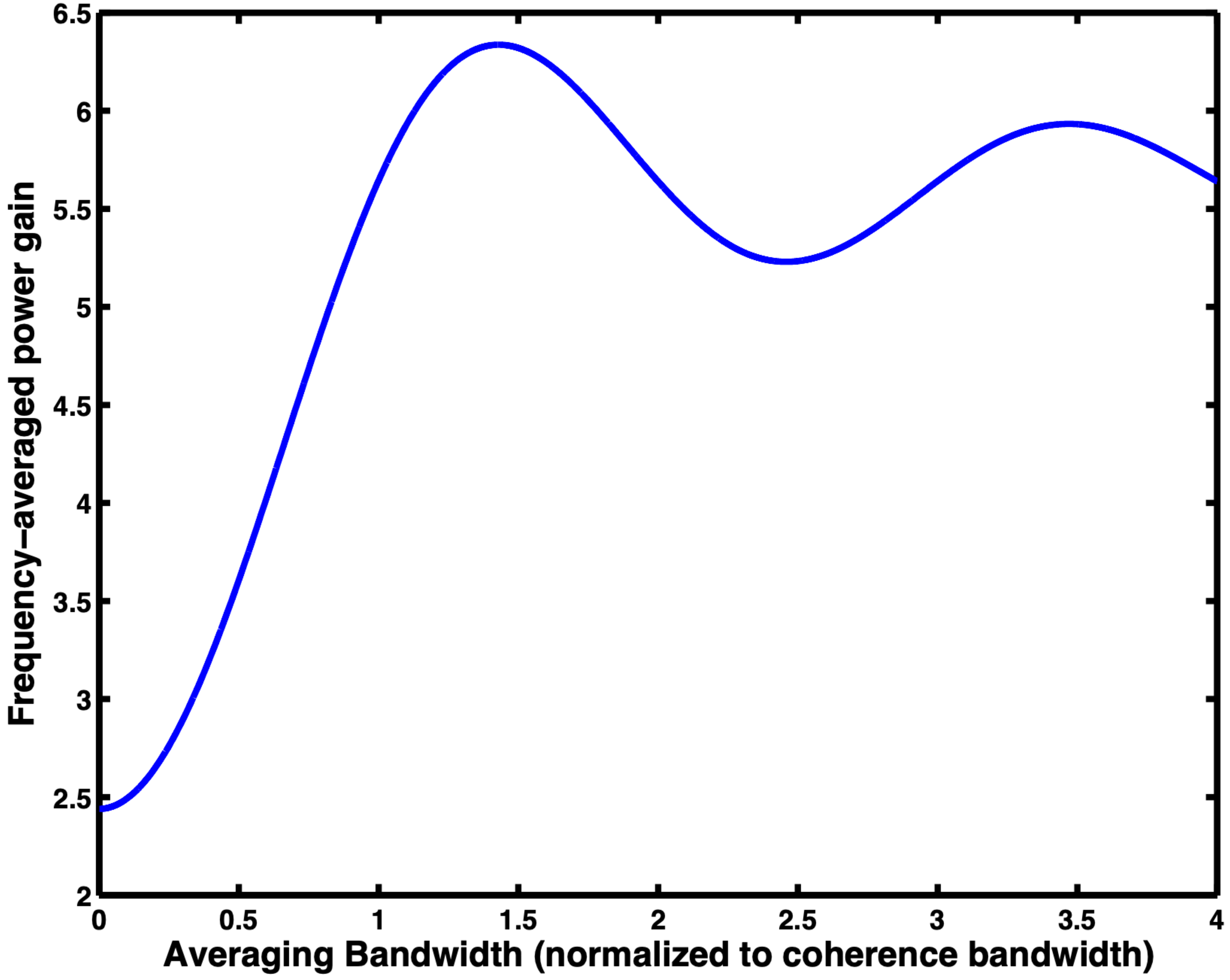
Average Channel Power Gain

$$\bar{G}(W) = \frac{1}{W} \int_{-W/2}^{W/2} |H(f)|^2 \, df$$

Average Channel Power Gain over a band $[-W/2, W/2]$

Plot $\bar{G}(W)$ as a function of W/B_c

Average Channel Power Gain



Amplitude Modulation

Problem 3.1 Figure 3.45 shows a signal obtained after amplitude modulation by a sinusoidal message. The carrier frequency is difficult to determine from the figure, and is not needed for answering the questions below.

(a) Find the modulation index.

(b) Find the signal power.

(c) Find the bandwidth of the AM signal.

$u(t) = (A_c + m(t)) \cos 2\pi f_c t$, $a_{mod} = \frac{A}{A_c}$, $m(t) = A \cos 2\pi f_m t$

a) $\left.\begin{array}{l} A_c + A = 30 \\ A_c - A = 10 \end{array}\right\}$ $A = 10$, $A_c = 20$

$\boxed{a_{mod} = 1/2}$

b) $\boxed{\overline{u^2} = \frac{A_c^2 + A^2}{2} = 250}$

c) Envelope has a period of $1/2$ ms, message bw = 2KHz

$\boxed{\text{AM BW} = 4\text{KHz}}$

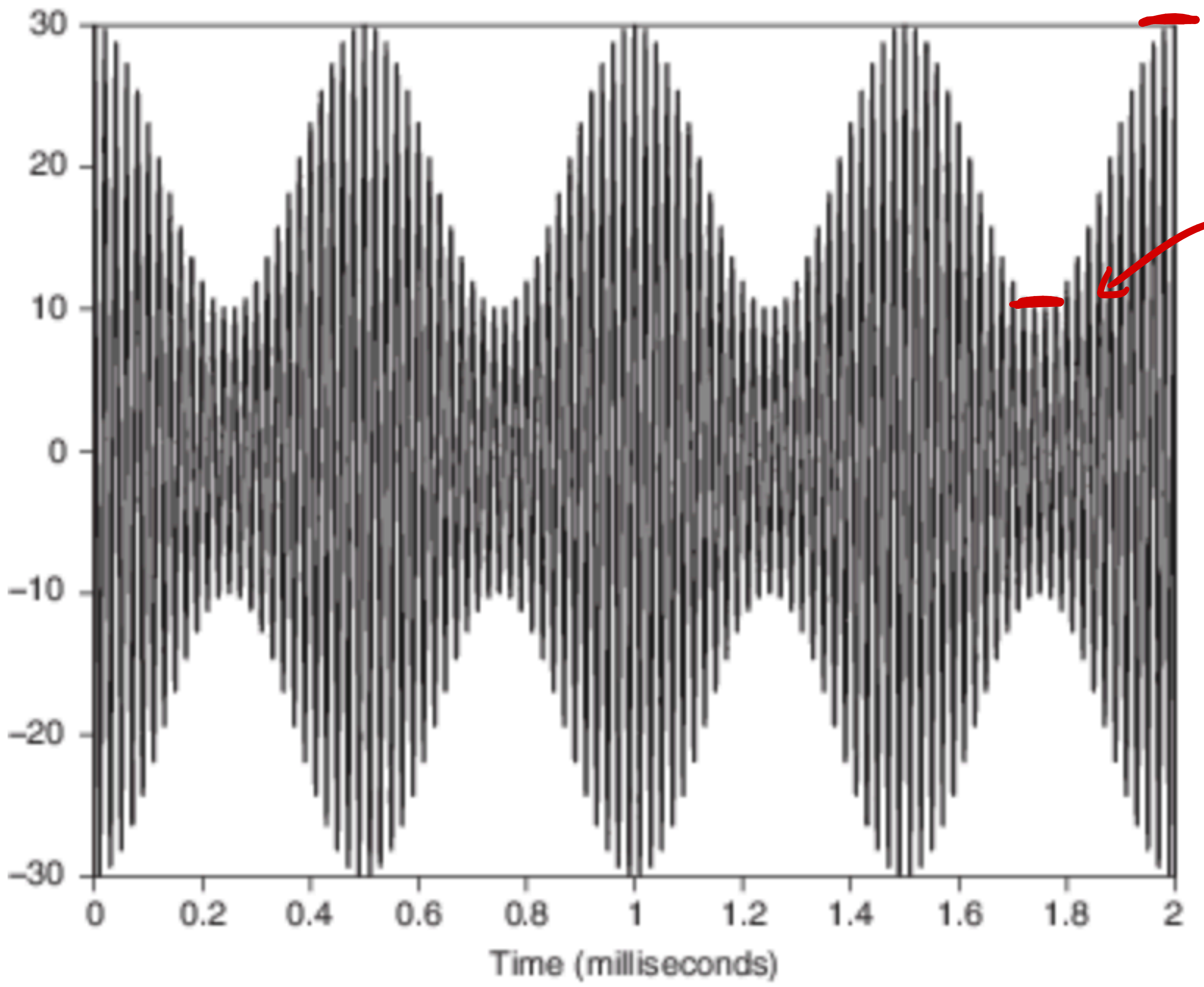


Figure 3.45 The amplitude-modulated signal for Problem 3.1.